

Sakshi Ravindra Asati

San Jose, CA | 303-356-2393 | sakshiasati51@gmail.com

linkedin.com/in/sakshiasati | github.com/sakshiasati17 | sakshiasati17.github.io/SakshiAsati-Portfolio/

Education

University of Colorado Boulder, Boulder, CO

Aug 2024 – May 2026

Master of Science in Data Science, GPA: 3.9/4.0

G H Raisoni College of Engineering, Nagpur, India

Aug 2018 – May 2022

Bachelor of Engineering in Information Technology, GPA: 3.8/4.0

Work Experience

Axion Orbital Space (YC W26), San Jose, CA (Remote)

Jul 2026 – Present

Machine Learning Engineer Intern

- Benchmarked 10 open-weight LLMs under consistent prompts and inference settings, measuring format adherence, numerical validity, reasoning leakage, latency, and failures to identify reliable serving behavior.
- Implemented a Python five-model failover workflow with retries, jittered backoff, rate-limit handling, request deadlines, concurrency controls, streaming-interruption detection, and latency/token telemetry; added numerical validation, four output guardrails, and SAR/ERA5 data fallbacks.

Cognizant Technology Solutions, Pune, India

May 2022 – May 2024

Programmer Analyst – Data Engineering

- Maintained production workflows processing 10M+ daily records from 5–6 sources, monitoring 100+ acceptance and production jobs across a 500+ workflow platform for availability, dependencies, SLAs, and data quality.
- Developed maintainable Python, SQL, and PySpark validation components for schema changes, nulls, duplicates, missing files, and source-to-target reconciliation before publishing partitioned Parquet data to HDFS, Hive, and AWS S3.
- Resolved software and data defects through log analysis, metrics examination, dependency tracing, root-cause investigation, and controlled reruns, helping reduce recurring production incidents by 50% across Spark failures, delayed feeds, connection errors, and failed loads.
- Supported Hadoop-to-AWS and scheduler-to-Airflow migrations, verified data and job behavior during rollout, and delivered unit-tested fixes through Bitbucket, Jenkins CI/CD, and Linux, contributing to a 30% reduction in workflow downtime.

Technical Skills

Programming & Fundamentals: Python, SQL, JavaScript, Pandas, dbt, Tableau, Plotly

Frameworks & AI/ML: FastAPI, REST APIs, API Integrations, SQLAlchemy, PyTorch, Scikit-learn, YOLOv8, NumPy, SciPy, Statsmodels

Data & Cloud: PostgreSQL, Qdrant, Redis/Celery, PySpark, Hadoop/HDFS, Hive, Airflow, AWS S3, GCP Cloud Run

Testing & Tools: Data Validation, Workflow Orchestration, Monitoring, Incident Resolution, pytest, Unit Testing, Debugging, Log Analysis, Docker, Nginx, Git, Bitbucket, Jenkins CI/CD, Linux

Projects

FaithTrace: RAG Evaluation & Diagnostics Platform

Jan – Apr 2026

- Designed an eight-service Docker application using async FastAPI, PostgreSQL, Qdrant, Celery/Redis, Nginx, and Next.js, with asynchronous task chains, metadata storage, vector search, API routing, health checks, and rate limiting.
- Evaluated 256 pipeline configurations across 301 real runs, uncovering 195 silent Ragas failures; recorded best faithfulness of 0.8875, context recall of 0.9000, p50 latency of 734.4 ms, and cost of \$0.0010, and validated diagnostics with 42 automated tests.

VisionStock: Retail Computer Vision Platform

Feb – Apr 2026

- Built a 13-route FastAPI/PostgreSQL application for image ingestion, product detection, planogram comparison, and inventory discrepancies, supported by a five-page Streamlit interface and containerized deployment to GCP Cloud Run.
- Fine-tuned YOLOv8n on 111 images across 34 classes, improving recall from 0% to 11.79% and mAP50 from 0% to 4.04%; measured 64.4 ms average and 77.4 ms p95 warm CPU inference and documented small-test-set and class-imbalance limitations.

StatForge: A/B Testing & Experimentation Platform

Jan – Mar 2026

- Developed seven reusable Python statistical modules with NumPy, SciPy, and Statsmodels and validated them through 43 unit and nine simulation tests, demonstrating approximately 5% Type-I error across 5,000 null experiments and 95% confidence-interval coverage.
- Built a PostgreSQL/SQLAlchemy data layer and five-model dbt pipeline with Streamlit/Plotly, Tableau exports, and HTML reports for 500K Criteo rows; measured 62.59% conversion lift and increased CUPED variance reduction from 1.84% to 24.74%.